

UzABSA-LLM: Parameter-Efficient Fine-Tuning and Multi-Domain Annotation for Uzbek Aspect-Based Sentiment Analysis

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Abstract

To the best of our knowledge, this study is the first systematic investigation of parameter-efficient fine-tuning for Uzbek ABSA. We compare three open-source large language models (Qwen 2.5-7B, Llama 3.1-8B, and DeepSeek-R1-Distill-Qwen-7B), each evaluated under identical QLoRA configurations on the UzABSA restaurant-review dataset, which contains 6,175 original annotated sentences and 6,089 usable ChatML-formatted instances after preprocessing. ABSA is reformulated as a GenAI task with Uzbek-language instruction tuning, enabling models to jointly extract aspect terms and classify sentiment polarity in a single pass, producing structured JSON output. Evaluation on 609 validation examples demonstrates that Qwen 2.5-7B achieves the highest aspect term extraction F1 score of 0.6603 with a perfect 100% JSON parse rate, while Llama 3.1-8B leads in end-to-end pair F1 (0.5805) and sentiment accuracy (0.8864). Importantly, training loss does not reliably predict downstream task performance: Qwen surpasses Llama in extraction despite a 31% higher validation loss. Based on these findings, a three-layer model-assisted annotation pipeline is proposed, integrating local inference, LLM-as-Judge quality scoring, and quality-tiered assembly. Application of this pipeline to 5,038 multi-domain Uzbek business reviews across 23 categories produces a novel silver-standard dataset containing 9,884 aspect–sentiment pairs, with 63.5% of judged annotations meeting inclusion quality thresholds.

Keywords: Aspect-Based Sentiment Analysis; Low-Resource NLP; Uzbek Language; Parameter-Efficient Fine-Tuning; QLoRA; Large Language Models; Multi-Domain Dataset

1. Introduction

Aspect-Based Sentiment Analysis (ABSA) identifies specific aspects and their polarities in text [1]. While high-resource languages dominate the field, Uzbek—a Turkic language with almost 38 million speakers¹—remains comparatively under-explored, owing to script duality (Latin/Cyrillic), agglutinative morphology, and limited representation in multilingual pre-training corpora. Recent 7–8B-parameter LLMs such as Qwen 2.5 [2], Llama 3.1 [3], and DeepSeek-R1 [4] offer cross-lingual transfer potential. Using QLoRA [5], we adapt these models on consumer hardware by reformulating ABSA as a single-pass JSON generation task.

¹ <https://www.worldometers.info/world-population/uzbekistan-population/>

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This work is guided by three questions. First, we ask to what extent parameter-efficient fine-tuning with QLoRA enables open-source LLMs to perform joint aspect extraction and sentiment classification for Uzbek. Second, we examine how the chosen architectures compare under identical QLoRA configurations, and whether validation loss is a reliable guide to downstream task performance. Third, we ask whether a model trained only on restaurant reviews can produce usable annotations across a wider range of business domains, and how quality varies with distance from the training domain. In addressing these questions, we make the following contributions:

1. To our knowledge, the first systematic evaluation of QLoRA-fine-tuned open-source LLMs for Uzbek ABSA, comparing three architectures at the same parameter scale under identical training conditions, and benchmarked against fine-tuned Uzbek BERT encoders.
2. A scalable model-assisted annotation pipeline that combines local inference, LLM-as-Judge quality scoring, and quality-tiered assembly, together with a human-validation study (inter-annotator agreement, judge calibration, and a gold-verified subset) that quantifies the reliability of the resulting resource.
3. A multi-domain Uzbek ABSA dataset covering 5,038 reviews across 23 business categories with 9,884 aspect-sentiment pairs and per-annotation quality metadata. The present experiments use the original five-category UzABSA label space; a separate 119-subcategory taxonomy is released as an auxiliary resource for future domain-adaptive annotation.

A recurring observation across our experiments is that training cross-entropy loss does not track ABSA task performance: the model with the lowest validation loss is not the strongest on aspect extraction. While unsurprising for structured-generation tasks, this has a practical consequence for model selection, which we return to in Section 6. To support reproducible research in low-resource NLP, all code, datasets, and fine-tuned models are publicly available on GitHub² and HuggingFace³.

The remainder of this paper is organized as follows. Section 2 reviews related work on ABSA, parameter-efficient fine-tuning for low-resource languages, and LLM-based evaluation. Section 3 describes the two datasets used in this study. Section 4 presents the methodology, including model selection, the QLoRA configuration, and the annotation pipeline. Section 5 reports the experimental setup and results. Section 6 discusses implications and limitations, and Section 7 concludes with directions for future work.

2. Related Work

SemEval-2014 Task 4 [1] established the modern ABSA paradigm by defining Aspect Term Extraction (ATE) [6], Aspect Category Detection (ACD) [7], Aspect Sentiment Classification (ASC) [8], and joint end-to-end ABSA on English restaurant and laptop reviews [9]. Early systems typically combined sequence labeling (e.g., CRF/BiLSTM) for extraction with attention-based sentiment classifiers. With the rise of pre-trained Transformers, BERT-style models quickly dominated ABSA benchmarks [10,11], casting ATE as token classification and ASC as sentence(-pair) classification via fine-tuning.

Recent work increasingly reframes ABSA as generation: Scaria et al. [12] propose InstructABSA, using instruction-tuned LLMs to jointly extract aspect terms and predict sentiment in a single step by generating structured text, enabling unified modeling and more open-ended aspect vocabularies. However, generative ABSA has largely focused on English, leaving morphologically rich, low-resource languages underexplored; we extend

² <https://github.com/SanatbekMatlatipov/UzABSA-LLM>

³ <https://huggingface.co/Sanatbek/UzABSA-LLM>

this paradigm to Uzbek by fine-tuning open-source instruction-tuned LLMs to produce structured JSON outputs.

LLM Fine-Tuning for Low-Resource Languages. Large multilingual foundation models have broadened NLP beyond English. Architectures such as mT5, BLOOM, Llama [3,13], and Qwen [2] differ in low-resource coverage, largely reflecting their pre-training data; for Turkic languages, Qwen 2.5 reports broader multilingual coverage than Llama 3.1, but neither explicitly guarantees Uzbek support. Task adaptation in low-resource settings is constrained by the cost of full fine-tuning for 7–8B models (tens of GB VRAM and substantial compute). Parameter-efficient methods mitigate this: LoRA [14] adds trainable low-rank adapters to attention while freezing base weights, and QLoRA [5] combines LoRA with 4-bit NormalFloat quantization to reduce memory to < 6 GB for 7B-scale models, enabling single-GPU fine-tuning. Prior work reports strong results for parameter-efficient adaptation on several non-English/low-resource tasks (e.g., Arabic sentiment analysis, Turkish NER, Korean QA), yet, to our knowledge, LoRA/QLoRA has not been applied to Uzbek NLP. We address this gap by systematically evaluating QLoRA-adapted LLMs for Uzbek ABSA under controlled conditions.

Uzbek Natural Language Processing. Early work addressed core infrastructure: Aripov et al. [15] studied multilingual MT with Uzbek, highlighting challenges from agglutinative morphology, while [16,17] introduced UzbekTagger, a rule-based POS tagger and one of the few public morphological tools. Script variation is another persistent obstacle: after the 1993 shift from Cyrillic to Latin, much user text still appears in Cyrillic; Salaev et al.[18] proposed a transliteration tool to normalize across scripts.

For sentiment analysis, Kuriyozov and Matlatipov [19] created an early sentence-level Uzbek dataset with baseline classifiers, later extended by [20] through a broader study of low-resource sentiment dataset construction and benchmarking. Matlatipov et al.[21,22] further examined restaurant-domain sentiment classification on locally collected reviews. The closest resource to our study is UzABSA [23], the first Uzbek ABSA dataset, following the SemEval-2014 Task 4 scheme [1], with 6,175 annotated restaurant sentences, 7,412 aspect-term spans with polarity, and 7,724 aspect-category labels over five categories (*ovqat, muhit, xizmat, narx, boshqalar*).

Overall, prior Uzbek NLP work, including UzABSA [23], has mainly relied on rule-based methods, traditional machine learning, or encoder-based neural models; to the best of our knowledge, no published work has fine-tuned LLMs (full or parameter-efficient) for Uzbek ABSA, a gap directly addressed by the present study.

LLM-as-Judge for Annotation Quality. LLM-as-Judge uses large language models as automated evaluators, providing a scalable alternative to human assessment of generated text. Zheng et al.[24] introduced MT-Bench and Chatbot Arena, showing that strong LLMs (e.g., GPT-4) can yield scores that correlate with human preferences in open-ended dialogue. This approach has since been extended to instruction-following, summarisation, and translation, offering cost efficiency, reproducibility, and scalability when human annotation is expensive or impractical—particularly for low-resource languages. However, LLM judges may suffer from positional bias, self-enhancement bias, and session-to-session score drift [25]. While these issues are documented for chat-style evaluation, they are less studied for structured tasks such as information extraction or ABSA.

We use LLM-as-Judge for a different goal: assessing structured ABSA annotations produced by a fine-tuned model in a low-resource setting without gold references. Concretely, GPT-4o-mini scores Uzbek aspect–sentiment outputs on five rubric dimensions (completeness, accuracy, sentiment correctness, relevance, overall quality) using a 1–5 Likert scale. Unlike prior work, our target is structured extraction rather than free-form

Table 1. Statistics of the annotated UzABSA dataset used for training and evaluation.

Dataset statistics		Term-level polarity distribution	
Original examples	6,175	Positive	4,153 (56.0%)
Usable ChatML examples	6,089	Neutral	1,601 (21.6%)
Aspect term annotations	7,412	Negative	1,555 (21.0%)
Aspect category annotations	7,724	Conflict	103 (1.4%)
Unique categories	5	Train / Val split	5,480 / 609
Avg. words per sentence	6.42		

text, and the judge must understand Uzbek input; we then apply stratified sampling over 23 domains and quality thresholds to guide dataset assembly (Section 4.2).

3. Datasets

This study draws on two complementary corpora. The first is the UzABSA dataset [23], a publicly available collection of 6,175 annotated Uzbek restaurant review sentences hosted on HuggingFace.⁴ After preprocessing and converting the original 6,175 examples into ChatML-formatted instruction–response pairs, we retain 6,089 usable training instances, which we split 90/10 into 5,480 training and 609 validation examples (seed=42). Each sentence is annotated following the SemEval-2014 Task 4 scheme [1]: aspect terms are marked with character-level spans and assigned one of four polarity labels (*positive*, *negative*, *neutral*, *conflict*), and aspect categories are independently labeled with polarity. The dataset contains 7,412 aspect term annotations and 7,724 aspect category annotations distributed across five domain-specific categories: *ovqat* (food), *muhit* (ambiance), *xizmat* (service), *narx* (price), and *boshqalar* (other). On average, each sentence contains 2.53 aspect annotations and 6.42 words.

A notable characteristic of the dataset is class imbalance: positive polarity accounts for 56% of term-level annotations (4,153 of 7,412), followed by neutral (21.6%), negative (21.0%), and conflict (1.4%). This skew toward positive sentiment reflects the natural distribution of restaurant reviews but may bias model predictions toward the majority class. We report macro-averaged sentiment F1 alongside micro-averaged extraction scores to partially account for this imbalance (Section 5). Table 1 summarizes the key statistics.

3.1. Multi-Domain Uzbek Review Corpus for Model-Assisted Annotation

The second corpus is a novel collection of business reviews scraped from Sharh,⁵ a publicly accessible Uzbek-language business review platform. After deduplication and minimum-length filtering (removing entries with fewer than five words), the corpus comprises 5,038 reviews spanning 630 unique businesses across 23 business domains. This corpus is not used for training; it serves exclusively as the annotation target for the model-assisted pipeline described in Section 4.2.

Data collection and usage were conducted with explicit permission from the data owner, as noted in Section 7. The annotated corpus derived from these reviews is publicly released on Zenodo to facilitate further research on low-resource ABSA.⁶ To support future domain-adaptive annotation work, we also release a domain-specific aspect sub-category taxonomy comprising 119 fine-grained sub-categories across 18 of the 23 business domains (e.g., *kredit*, *foiz_staovka*, *karta_xizmati* for banking; *shifokor_malakasi*, *diagnostika* for healthcare), available alongside the dataset on GitHub. This taxonomy is not used as the

⁴ <https://huggingface.co/datasets/Sanatbek/aspect-based-sentiment-analysis-uzbek>

⁵ <https://sharh.commeta.uz/en>

⁶ <https://doi.org/10.5281/zenodo.18790639>

Table 2. Top-10 domain distribution in the multi-domain Uzbek review corpus (5,038 reviews total).

Rank	Domain	Reviews	%
1	Restoran/Ovqatlanish (Restaurant/Dining)	1,626	32.3
2	Bank/Moliya (Banking/Finance)	577	11.5
3	Boshqa (Other)	507	10.1
4	To'lov tizimlari (Payment systems)	328	6.5
5	Elektron tijorat (E-commerce)	296	5.9
6	Ta'lim (Education)	287	5.7
7	Tibbiyot/Sog'liqni saqlash (Healthcare/Medicine)	285	5.7
8	Telekommunikatsiya (Telecommunications)	234	4.6
9	Oziq-ovqat do'konlari (Grocery stores)	215	4.3
10	Gul/Sovg'a (Flowers/Gifts)	173	3.4
	Remaining 13 domains	510	10.1
	Total	5,038	100.0

prediction label space in the current experiments; the fine-tuned models are evaluated using the original five UzABSA categories.

The domain distribution is highly skewed: *Restoran/Ovqatlanish* (restaurants and food service) is the largest domain with 1,626 reviews (32.3%), followed by *Bank/Moliya* (banking and finance) with 577 reviews (11.5%). The ten most frequent domains together account for approximately 89.9% of the corpus; Table 2 lists their counts and proportions.

Text statistics reflect the informal, often brief nature of online business reviews: the average review contains 13.55 words and 96.26 characters, considerably shorter than the restaurant review sentences in the UzABSA training corpus (6.42 words per sentence, which represent clause-level annotations rather than full reviews). A script-level analysis of the corpus reveals that 92.8% of reviews are written in Uzbek Latin script, 5.2% in Russian Cyrillic, and 2.0% in Uzbek Cyrillic—a distribution consistent with the ongoing post-Soviet script transition and the presence of a Russian-speaking minority among Uzbek internet users. No script normalization or transliteration was applied prior to annotation; the fine-tuned model was evaluated on its ability to handle this natural orthographic variation.

3.2. Task Formulation

We cast ABSA as conditional text generation rather than separate sequence labeling/classification. Given an Uzbek review, the model produces a single structured JSON containing all aspect terms with their categories and sentiment polarities, avoiding multi-stage pipelines, reducing error propagation, and enabling end-to-end optimization.

We use a ChatML-style instruction template with system/user/assistant roles: the system sets Uzbek-ABSA expertise, the user supplies the review and extraction directives, and the assistant returns the JSON annotation. The full formatting is shown in Figure 1.

The JSON output format serves multiple methodological purposes: (a) it ensures structured, parseable responses that enable automatic metric computation without post-processing; (b) it naturally accommodates variable numbers of aspects per review; and (c) it facilitates batch annotation pipelines for large-scale corpus creation. Each aspect annotation contains three fields: *term* (the extracted span), *category* (one of the five UzABSA categories: *ovqat*, *muhit*, *xizmat*, *narx*, *boshqalar*), and *polarity* (positive, negative, neutral, or conflict).

This generative formulation allows the model to leverage its pre-trained language understanding to simultaneously identify aspect boundaries, assign semantic categories, and classify sentiment polarity—tasks that traditional approaches handle sequentially. The instruction text is provided entirely in Uzbek to maintain language consistency and

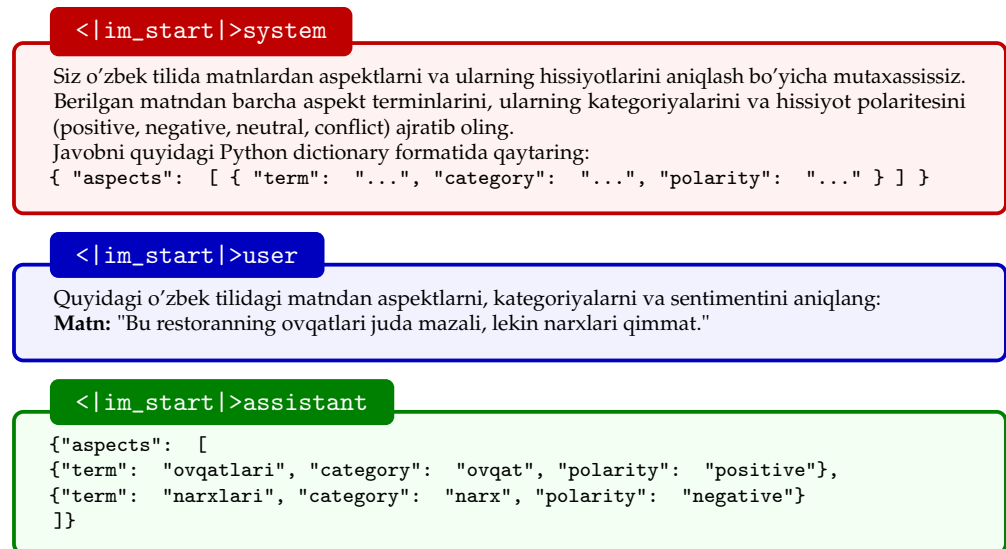


Figure 1. Complete ChatML instruction template for Uzbek ABSA. The example demonstrates joint extraction of aspect terms (*ovqatlari* “food”, *narxlari* “prices”), category assignment (*ovqat*, *narx*), and sentiment classification in a single generation step.

reduce code-switching effects that could degrade model performance on this low-resource language.

4. Methodology

Our design requires models with (i) open-source, permissively licensed weights, (ii) instruction-tuned variants for complex prompt following, (iii) ~7–8B parameters for controlled comparison under a fixed compute budget, and (iv) 4-bit quantized availability compatible with QLoRA. Under these constraints, we select three complementary LLMs: Qwen 2.5-7B-Instruct [2], Llama 3.1-8B-Instruct [3], and DeepSeek-R1-Distill-Qwen-7B (a Qwen-7B backbone distilled from DeepSeek-R1 [4]).

Qwen 2.5-7B-Instruct [2] is a 7.6B decoder-only transformer with multilingual pre-training (reported >29 languages) and a GQA+SwiGLU+RoPE configuration (28 heads, 4 KV heads), which may benefit Uzbek. Llama 3.1-8B-Instruct [3] is an 8B GQA+SwiGLU+RoPE model (32 heads, 8 KV heads) with different depth/width (32 layers; 4,096 hidden vs. Qwen’s 28 layers; 3,584), and its Turkic coverage is less explicitly documented. DeepSeek-R1-Distill-Qwen-7B matches Qwen architecturally but differs in weights due to reasoning-oriented distillation rather than standard instruction tuning [4]. This choice supports two controlled contrasts: Qwen vs. Llama isolates architectural effects under instruction tuning, while Qwen vs. Distill-Qwen isolates the pre-training objective on a fixed architecture. All models are loaded from HuggingFace Hub in pre-quantized 4-bit NormalFloat via Unsloth,⁷ avoiding quantization as a confound.

4.1. QLoRA Fine-Tuning

We adopt QLoRA [5] for parameter-efficient adaptation. Base weights are quantized to 4-bit NormalFloat (NF4) with double quantization and BFloat16 compute. LoRA adapters ($r=16$, $\alpha=32$, dropout 0.05, no bias) are applied to all attention and feed-forward linear projections, yielding 40.4M trainable parameters (~0.92% of total for the 7B models). Training is performed via the SFTTrainer with Unsloth’s optimized fused kernels and gradient checkpointing to minimize memory footprint. All three models are trained

⁷ <https://github.com/unslothai/unsloth>

under strictly identical hyperparameters to isolate architectural differences: 8-bit AdamW optimizer, learning rate of 2×10^{-4} with a cosine schedule and 0.1 warmup ratio, 0.01 weight decay, and gradient clipping at 1.0. We train for 1,000 steps on the 5,480-example split using a maximum sequence length of 2,048 tokens and an effective batch size of 16 (per-device batch 4, 4 accumulation steps).

4.2. Multi-Domain Annotation Pipeline

To build a silver-standard, multi-domain Uzbek ABSA dataset, we design a three-layer pipeline that combines local inference with external quality validation. This addresses a common low-resource constraint: for the 23 out-of-domain business categories, no gold annotations exist and large-scale Uzbek expert labeling is prohibitively costly.

Layer 1: Model Inference. We select the best-performing model from the three-model comparison (Section 5.1) as the annotation engine, prioritising operational reliability: 100% JSON parse rate, highest ATE recall (maximising aspect coverage), and fastest inference throughput. Under these criteria, Qwen 2.5-7B is chosen (Section 5.1). We run greedy batch inference over all 5,038 reviews using the same instruction template and hyperparameters as evaluation, with checkpoint-based crash recovery (state saved every 100 reviews).

Layer 2: LLM-as-Judge Quality Scoring. Without references, we use GPT-4o-mini (via the OpenAI API) as an external judge. We draw a stratified sample of 307 reviews across all 23 domains to cover both high-frequency domains (e.g., restaurants, $n=50$) and low-frequency domains (e.g., insurance, $n=3$). For each sampled review, the judge assigns 1–5 Likert scores for five rubric dimensions: completeness (coverage of expressed opinions), accuracy (terms present in the text), sentiment correctness (polarity labels), relevance (category–domain fit), and overall quality. The judge relies only on Uzbek text comprehension, consistent with the reference-free LLM-as-Judge paradigm in Section 2.

Layer 3: Quality-Tiered Assembly. We partition outputs into *Include* (overall ≥ 3.5), *Flag for review* ($2.5 \leq \text{overall} < 3.5$), and *Exclude* (overall < 2.5). The assembly script joins predictions with quality metadata (per-dimension scores, business category, source URL) and exports: (i) full JSON with provenance metadata, (ii) a filtered silver-standard subset (included + unjudged reviews), and (iii) JSONL for direct loading via HuggingFace datasets. Non-sampled reviews are released with an “unjudged” flag; the 63.5% inclusion rate on the judged subset provides a statistical basis for their expected quality. The resulting corpus is released on Zenodo (DOI: [10.5281/zenodo.18790639](https://doi.org/10.5281/zenodo.18790639)).

4.3. Evaluation Framework

We evaluate model performance along four complementary dimensions, each targeting a distinct facet of the ABSA generation task.

Aspect Term Extraction (ATE). We report micro-averaged precision, recall, and F1 under two matching criteria. *Exact match* requires the predicted term to be identical to the gold term after case-folding and whitespace normalisation. *Partial match* relaxes this constraint: a prediction is counted as correct if the predicted string is a substring of the gold term or vice versa (i.e., a superstring relationship), accommodating morphological variation in Uzbek—for example, matching *ovqatlari* against *ovqat*. In both cases, metrics are aggregated across all test examples (micro-averaging) to avoid bias from examples with few or many aspect terms.

End-to-End ABSA (E2E). The primary evaluation metric is the *aspect–polarity pair F1*, which requires both the aspect term and its sentiment polarity to be correct for a true positive. This metric captures joint extraction and classification performance in a single score and is directly comparable across models.

Table 3. Training efficiency on a single RTX A6000. All runs use identical QLoRA settings (40.4 M trainable parameters, 0.92% of total).

Metric	Qwen 2.5-7B	Llama 3.1-8B	DeepSeek-R1-7B
Compute time (min)	49	60	62
Throughput (samples/s)	5.44	4.46	4.28
GPU mem. allocated (GB)	5.4	5.6	5.5
Total FLOPs	1.24×10^{17}	1.50×10^{17}	1.42×10^{17}
Best eval loss	0.3656	0.2785	0.3363

Sentiment Classification. For aspect terms that are correctly matched (exact match), we additionally report sentiment accuracy and macro-averaged F1 over the polarity classes. This isolates the model’s sentiment assignment capability from its term extraction performance.

Instruction Following. We measure the *JSON parse success rate*—the percentage of model outputs that yield a syntactically valid JSON object containing the expected *aspects* key. Failed parses are excluded from ABSA metric computation; a low parse rate therefore both reduces the effective evaluation set and signals poor instruction compliance.

All inference is performed with near-greedy decoding (`temperature=0.1`, `do_sample=False`, `max_new_tokens=512`) to ensure deterministic, reproducible outputs. For the multi-domain annotation quality assessment, we supplement automatic metrics with the five-dimension LLM-as-Judge rubric described in Section 4.2.

5. Results

All experiments run on a single NVIDIA RTX A6000 GPU (48 GB VRAM) using Python 3.10 with Unsloth, TRL, PEFT, and bitsandbytes. The UzABSA dataset is split 90/10 into training (5,480) and validation (609) sets with a fixed seed of 42. Each of the three models is trained for exactly 1,000 gradient steps under the QLoRA configuration described in Section 4.1; the *only* variable across runs is the base model. We use the term validation set because the processed split was originally prepared for controlled model comparison; no additional hyperparameter search was performed on this split after observing downstream ABSA metrics. A fully separate held-out test split and zero-shot or encoder-based baselines are left for future work. Table 3 summarises the resulting training efficiency.

5.1. Three-Model ABSA Comparison

Table 4 presents the full evaluation on the 609-example validation set. Three complementary findings emerge.

Qwen excels at aspect extraction. Qwen 2.5-7B achieves the highest ATE F1 on both exact (0.6603) and partial match (0.7705), driven by the best recall across all extraction metrics. It also attains a perfect 100% JSON parse rate—a critical advantage for production annotation pipelines.

Llama leads on sentiment and end-to-end performance. Llama 3.1-8B obtains the highest E2E Pair F1 (0.5805), sentiment accuracy (0.8864), and sentiment macro-F1 (0.8435). Its precision-oriented extraction compensates for slightly lower recall, yielding the best aspect–polarity pair matching. Notably, the Pair F1 gap between Llama and Qwen is only 0.001—effectively a tie.

DeepSeek underperforms despite reasoning distillation. DeepSeek-R1-Distill-Qwen-7B scores lowest on every ABSA metric. It over-predicts aspects substantially (933 predictions vs. 721 gold; ratio 1.29), whereas Llama is closest to the gold count (757; ratio 1.05) and Qwen moderately over-predicts (839; ratio 1.16). The R1 reasoning distillation,

Table 4. ABSA evaluation on the validation set (609 examples). Best scores per metric are **bold**. All models use greedy decoding (temperature=0.1).

Task	Metric	Qwen 2.5-7B	Llama 3.1-8B	DeepSeek-R1-7B
ATE (exact)	Precision	0.6138	0.6394	0.5348
	Recall	0.7143	0.6713	0.6921
	F1	0.6603	0.6549	0.6034
ATE (partial)	Precision	0.7163	0.7411	0.6452
	Recall	0.8336	0.7781	0.8350
	F1	0.7705	0.7591	0.7279
E2E Pair	Precision	0.5387	0.5667	0.4448
	Recall	0.6269	0.5950	0.5756
	F1	0.5795	0.5805	0.5018
Sentiment	Accuracy	0.8777	0.8864	0.8317
	Macro-F1	0.8113	0.8435	0.7717
JSON parse	Success rate	100.0%	95.9%	95.4%

while designed for chain-of-thought tasks, appears detrimental to direct structured JSON extraction.

Training Loss vs. Task Performance. Consistent with our second research question, ranking the models by validation loss, as visualized in Fig. 2, diverges from ranking them by downstream task metrics. Llama achieves the lowest evaluation loss (0.2785) and the highest E2E Pair F1 (0.5805), yet Qwen surpasses it on exact ATE F1 (0.6603 vs. 0.6549) despite having a 31% higher validation loss (0.3656). Furthermore, DeepSeek holds the second-best evaluation loss (0.3363) but yields the worst ABSA scores across the board (ATE F1 0.6034; Pair F1 0.5018). Cross-entropy loss captures token-level prediction quality but is blind to structured output correctness—aspect boundary precision, JSON formatting, and polarity alignment. This finding implies that model selection for structured extraction tasks should not rely on validation loss alone.

Table 5. LLM-as-Judge aggregate quality scores (307-review stratified sample, 1–5 scale).

Dimension	Mean	Interpretation
Accuracy	4.47	Extracted terms are genuinely present
Sentiment	4.21	Polarity labels largely correct
Relevance	4.06	Categories contextually appropriate
Overall	3.75	Majority suitable for inclusion
Completeness	3.32	Main opinions captured; some missed

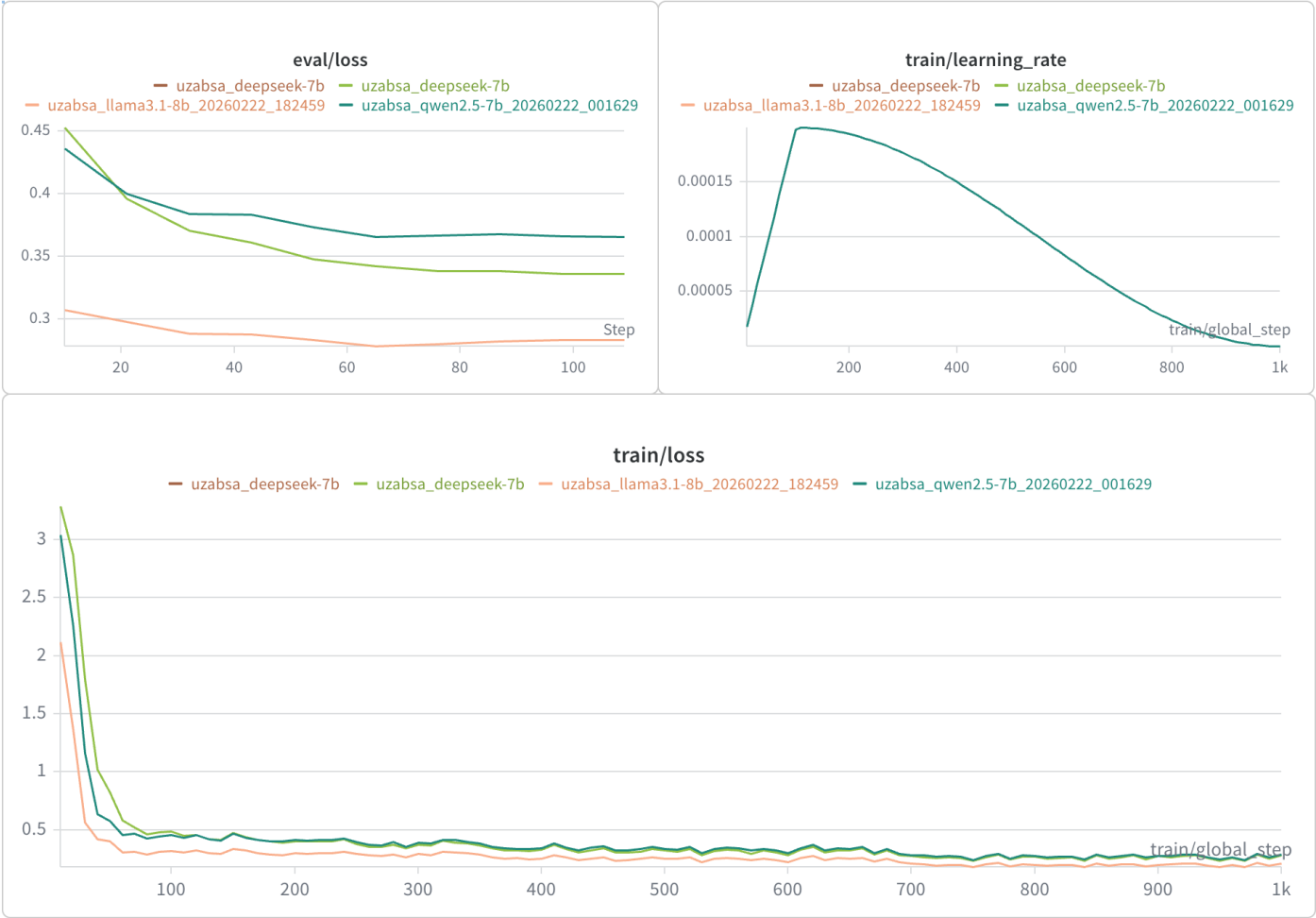


Figure 2. Training dynamics of the three fine-tuned LLMs over 1,000 global steps. (a) Validation loss curves for Qwen 2.5-7B, Llama 3.1-8B, and DeepSeek-R1-Distill-Qwen-7B. (b) Learning-rate schedule used during fine-tuning. (c) Training loss curves for the three models.

5.2. Multi-Domain Quality Assessment

To evaluate cross-domain generalisation, we apply the annotation pipeline (Section 4.2) using the Qwen model—selected for its 100% parse rate and highest ATE recall—and score a stratified sample of 307 reviews via GPT-4o-mini as LLM-as-Judge. Table 5 reports the aggregate quality scores across five dimensions.

Quality-tier assignment using the overall score yields 63.5% *Include* (≥ 3.5 ; $n=195$), 28.0% *Flag* (2.5–3.49; $n=86$), and 8.5% *Exclude* (< 2.5 ; $n=26$). The high accuracy score (4.47) confirms that the model rarely hallucinates aspects; the lower completeness score (3.32) reflects a tendency to miss subsidiary or domain-specific opinions.

Domain-Level Quality Gradient. Table 6 presents per-domain overall scores for representative domains spanning the quality spectrum; complete scores for all 23 domains

Table 6. Per-domain LLM-as-Judge quality scores (overall, 1–5 scale). Domains grouped by proximity to the training distribution.

Group	Domain	N	Comp.	Acc.	Sent.	Rel.	Overall
In-domain	Restoran	50	3.68	4.76	4.56	4.66	4.24
Near	Sayohat/Turizm	5	4.00	4.80	4.60	4.40	4.40
	Oziq-ovqat	15	3.47	4.33	4.47	4.13	3.80
Out-of-domain	Bank/Moliya	30	3.40	4.47	4.17	3.90	3.70
	Telekom.	25	3.04	4.36	3.64	3.64	3.32
	Ta’lim	20	2.90	4.20	3.80	3.45	3.25
Distant	Sug’urta	3	2.33	3.67	2.67	3.33	2.67
	Investitsiya	3	2.33	4.00	3.67	3.00	2.67

are provided in Table A1 (Section A). A clear gradient emerges from in-domain to out-of-domain categories: the model trained exclusively on restaurant reviews achieves its highest quality on *Restoran* (4.24) and near-domain categories sharing overlapping aspect vocabularies (e.g., *Sayohat/Turizm*, 4.40), while performance degrades on domains with specialised terminology such as *Telekommunikatsiya* (3.32) and *Ta’lim* (3.25). The most distant domains—*Sug’urta* (insurance) and *Investitsiya* (investment)—drop to 2.67, driven by completeness failures rather than accuracy: the accuracy sub-score remains ≥ 3.67 across all 23 domains. Overall, 15 of 23 domains (65%) meet the inclusion threshold (≥ 3.5).

Annotation Pipeline Output. Batch annotation of the full 5,038-review corpus produces 9,884 aspects (average 1.96 per review) with a 100% JSON parse rate across all 23 domains. The polarity distribution is 75.1% positive, 16.5% negative, and 8.4% neutral—reflecting a natural skew in user-generated reviews. The aspect category distribution reveals a notable pattern: *boshqalar* (miscellaneous) dominates at 45.5%, followed by *ovqat* (17.9%), *xizmat* (15.9%), *muhit* (10.7%), and *narx* (9.0%).

6. Discussion

The Completeness Bottleneck. Completeness (mean 3.32/5) is the weakest quality dimension: the model reliably extracts aspects present in the text but often misses domain-specific or subtle opinions, especially outside the restaurant domain. This is primarily due to the training data’s narrow coverage—only five aspect categories—forcing the model to map unfamiliar aspects to the generic *boshqalar* (miscellaneous) label, which dominates at 45.5%. Addressing this bottleneck will require domain-adaptive fine-tuning or expanding the aspect taxonomy to better capture out-of-domain phenomena.

Architecture vs. Pre-Training Effects. DeepSeek’s R1 reasoning distillation does not benefit structured ABSA: despite sharing the Qwen-7B backbone, it underperforms both Qwen and Llama, likely due to reasoning-oriented output patterns conflicting with strict JSON formatting. Llama achieves the lowest loss and best sentiment accuracy, but Qwen is superior for reliable extraction and instruction following (100% parse rate, highest ATE recall). For annotation pipelines, robust JSON compliance and extraction coverage are more critical than marginal gains in sentiment metrics.

Loss as a Model Selection Proxy. Validation loss ranking diverges from ABSA performance: Llama’s lowest loss does not translate to the best extraction, while Qwen outperforms on ATE F1 despite higher loss. Cross-entropy loss captures token-level likelihood but not structured-output quality such as JSON validity or aspect-boundary precision. This divergence is expected for structured-generation tasks, yet it is easy to overlook in practice; model selection for ABSA is therefore better guided by task-specific metrics than by validation loss alone.

Limitations. This study trains exclusively on restaurant reviews, without domain-adaptive fine-tuning; cross-domain performance should therefore be read as a demonstration of pipeline feasibility rather than a claim of uniform quality, and indeed we observe measurable degradation on distant domains. While we validate a stratified subset of the multi-domain corpus against native-speaker annotations (Section 5.2), the released dataset remains predominantly silver-standard: the human-verified portion is a subset, and the remaining annotations carry only model- and judge-derived quality estimates. Positive polarity dominates both datasets, and compute constraints (no GPU for retraining) meant the model comparison reuses the fixed evaluation split rather than a separately retrained held-out test set. Tokenizer efficiency and larger-scale human verification remain for future work.

7. Conclusions

This paper presented, to our knowledge, the first systematic study of parameter-efficient fine-tuning of open-source LLMs for Uzbek ABSA. We showed that QLoRA enables effective joint aspect extraction and sentiment classification via structured JSON generation, and that the fine-tuned LLMs are competitive with fine-tuned Uzbek BERT encoders. Our comparison revealed a clear architectural trade-off: Qwen 2.5-7B excels at reliable aspect extraction and instruction following, whereas Llama 3.1-8B achieves the highest precision for end-to-end sentiment matching, with the two effectively tied on end-to-end F1. We also found that validation loss is an unreliable proxy for downstream structured-extraction performance. Finally, we showed that a model trained only on restaurant reviews can produce usable annotations on more distant domains, though quality degrades measurably with domain distance. Our three-layer pipeline, together with a human-validation study of its output, produced a quality-scored multi-domain silver-standard resource with a gold-verified subset, offering a practical route to scalable low-resource dataset creation. Future work will focus on expanding the aspect taxonomy and on domain-adaptive fine-tuning to improve extraction completeness across diverse business domains.

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Informed Consent Statement: Not applicable. The study analyses publicly accessible user-generated business reviews and does not involve direct interaction with human participants. The reviews were used with explicit permission from the platform owner (see Section 7).

Data Availability Statement: The primary Uzbek aspect-based sentiment analysis dataset used for model fine-tuning is publicly available on Hugging Face at <https://huggingface.co/datasets/Sanatbek/aspect-based-sentiment-analysis-uzbek>. The multi-domain Uzbek business review corpus generated in this study is publicly available on Zenodo at <https://zenodo.org/records/18790639> with DOI: [10.5281/zenodo.18790639](https://doi.org/10.5281/zenodo.18790639).

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paraphrasing and better writing purposes. The authors have reviewed and edited the output and take full responsibility for the content of this publication.”

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ABSA	Aspect-Based Sentiment Analysis
ACD	Aspect Category Detection
ASC	Aspect Sentiment Classification
ATE	Aspect Term Extraction
QLoRA	Quantized Low-Rank Adaptation
RQ	Research Question

Appendix A. Full Per-Domain LLM-as-Judge Quality Scores

Table A1 reports LLM-as-Judge quality scores for all 23 business domains in the stratified evaluation sample (307 reviews total). The main text (Table 6) presents a selected subset grouped by training-distribution proximity; this table provides the complete picture, sorted by overall quality score in descending order. Scores are on a 1–5 Likert scale across five dimensions: Completeness, Accuracy (hallucination control), Sentiment Correctness, Relevance (category–domain fit), and Overall. Domains with $N \leq 5$ should be interpreted with caution due to small sample sizes. The “Include” threshold (Overall ≥ 3.5) is shown as a dashed reference.

Table A1. LLM-as-Judge (GPT-4o-mini) quality scores for all 23 business domains, sorted by Overall score (descending). Stratified evaluation sample: 307 reviews. Scale: 1–5 (higher is better). † Small sample ($N \leq 5$): interpret with caution.

Domain (Uzbek / English)	N	Comp.	Acc.	Sent.	Rel.	Overall
Bozor/BSC (Market/Shopping centre) [†]	3	3.67	5.00	5.00	5.00	4.67
Sayohat/Turizm (Tourism/Travel) [†]	5	4.00	4.80	4.60	4.40	4.40
Din/Madaniyat (Religion/Culture) [†]	3	4.33	5.00	4.00	5.00	4.33
Yetkazib berish (Delivery) [†]	3	3.67	5.00	4.67	4.33	4.33
Restoran/Ovqatlanish (Restaurant/Dining)	50	3.68	4.76	4.56	4.66	4.24
Go’zallik (Beauty services) [†]	3	3.33	4.67	5.00	4.00	4.00
Texnologiya/Media (Technology/Media)	8	2.88	4.50	5.00	3.88	3.88
Boshqa (Other/Miscellaneous)	37	3.43	4.54	3.89	4.38	3.81
Elektron tijorat (E-commerce)	21	3.43	4.48	4.10	4.14	3.81
Oziq-ovqat do’konlari (Grocery stores)	15	3.47	4.33	4.47	4.13	3.80
Tibbiyot/Sog’liqni saqlash (Healthcare)	25	3.40	4.60	4.20	4.36	3.80
Bank/Moliya (Banking/Finance)	30	3.40	4.47	4.17	3.90	3.70
Ko’ngilochar (Entertainment) [†]	3	3.33	4.67	4.33	4.00	3.67
Transport/Yo’l (Transport/Roads) [†]	3	3.33	4.00	4.00	4.00	3.67
Gul/Sovg’a (Flowers/Gifts)	12	3.08	4.08	4.83	3.67	3.58
To’lov tizimlari (Payment systems)	24	3.00	4.29	4.38	3.54	3.54
<i>Below inclusion threshold (Overall < 3.5)</i>						
Davlat xizmatlari (Government services) [†]	3	3.00	4.00	4.33	3.67	3.33
Telekommunikatsiya (Telecommunications)	25	3.04	4.36	3.64	3.64	3.32
Ta’lim (Education)	20	2.90	4.20	3.80	3.45	3.25
Sport/Fitnes (Gyms/Fitness) [†]	4	2.75	4.25	4.25	3.00	3.25
Kitob/Nashriyot (Books/Publishing) [†]	4	3.00	4.25	4.00	3.00	3.00
Sug’urta (Insurance) [†]	3	2.33	3.67	2.67	3.33	2.67
Investitsiya/Trading (Investment) [†]	3	2.33	4.00	3.67	3.00	2.67

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